



Headphone Zone

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Jetsons movie

Jetsons - the classic cartoon series by Hannah-Barbera is being made into a movie http://dgit.in/Jetsons_Mov

Bazaar

Gigabyte GeForce GTX 960 G1 Gaming

Has the latest from NVIDIA hit the sweet spot?

Gigabyte's GeForce GTX 960 G1 Gaming graphics card heralds the arrival of the much awaited 60-class graphics card. The 60-class graphics cards offer the highest price to performance ratio while maintaining playable frame rates. Moreover, in India they've been priced around the ₹15,000 - ₹20,000 range so these cards sell good but not at launch. Gamers would rather wait for roughly a year or so before getting these cards and by then the price comes down to the more affordable ₹13,000 - ₹15,000 range. So that makes the 60-class cards the go-to card for every budget gaming PC build.

So when you take a look at the GM206 you see that it has exactly half the amount of CUDA cores, TMUs and ROPs than the GM204. Something that's gotten a little controversial since the 960's specifications leaked out on the Internet is that of the bus-width for the memory. The Maxwell architecture is one that scales really well and NVIDIA has opted for using compression algorithms to squeeze more data through the data-bus rather than increasing the bus-width. What adds fuel to the fire is that the competition sports a 256-bit bus width on R9 285 and even the GTX 760 has a 256-bit bus width. The question that needs to be answered is if the compression is so good that a 50% reduction in bandwidth is justified? Well we need to look at what we're getting and how much we're paying for it. In the case of the 760, it used the same chip as that of the 770 so the bus-width was unlikely to be that different from the 770. In the case of the 960, it isn't based on the same GM204 chip which the 970 is based on so it has been redesigned for the 960 to not only provide a comparable framerate but also by doing so at a much lower price point.

The TDP of the GM206 chip is 120W which is obviously not going to scale linearly but with such a low TDP we've seen silent cooling slowly become mainstream. Gigabyte's GTX 960 G1 Gaming card has ramped up the base clock from 1127MHz to 1241MHz and the boost clock has been increased from 1178MHz to 1304MHz. This massive overclock, which has been made thanks to Maxwell, requires a lot more power and for this, Gigabyte changed the reference design's 120W TDP through a three-phase VRM to a 150W six-phase VRM. A change that not only allow more headroom for



Price
₹19,900

overclocking but is also cooler since the power draw will be distributed across six-phases instead of three.

The G1 Gaming SKU has a pretty high factory overclock and for that they've come up with a good cooling solution. The front chassis or fan shroud is made out of aluminium and they've made use of four heat pipes that run across the entire expanse of the card's front. These heat pipes sit directly on top of the GPU and even the VRM has a block of aluminium to cool it down. On the rear, we have a back-plate which is also made of aluminium that spans across the entire PCB. For the G1 Gaming card the Ultra Durable technology translates to having two ounces of copper in their PCB along with high quality solid state capacitors and metal chokes.

Performance wise the GTX 960 has followed in the footsteps of the GTX 980. It is a much efficient card when it comes to power consumption and also stays cooler but it barely outperforms a factory overclocked GTX 760. However, we've already discussed how the 960 is a redesign that is suited for the mid-range segment and has launched at a much lower price point so expecting the linear performance scale akin to what was achieved from the 760 (in comparison to the 770) is unfair. So if you're still running a rig with a Fermi or Kepler 1st generation card then the GTX 960 is a worthwhile upgrade. The Gigabyte GeForce GTX 960 G1 Gaming card has a considerable improvement over the reference design and the factory overclock makes it worth the investment. We even managed to push the card a little further by overclocking it by an additional 90MHz and they claim that the card can go all the way up to 1379MHz. We'd say that buying two of these and rigging them together in SLI is a much better proposition than getting a single GTX 980.

Mithun Mobandas



Performance..... 47
Value 85
Build 69

Specifications

Chipset: GM206; **Base clock:** 1241MHz; **Memory clock:** 1753MHz; **Stream processors:** 1024; **Texture Units:** 64; **ROPs:** 32; **Manufacturing process:** 28nm, PCIe 3.0, 4096 x 2160 digital resolution support, 2 GB Memory; **DirectX support:** 12; **OpenGL support:** 4.4; **Power Connectors:** 6Pin + 6Pin; **TDP:** 150W

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